BRNO UNIVERSITY OF TECHNOLOGY FACULTY OF ELECTRICAL ENGINEERING AND COMMUNICATION

Robotics and AI

# MPC-MAP Assignment No. 4 - Report

**Author:** Tomáš Špelda

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## Task 1

Loaded the outdoor_1 map. The trajectory is defined by a spline-based function from start to finish with few control points.

The GNSS covariance matrix was initialized using historical GNSS data.

```
Inicializace GNSS
Změřená počáteční pozice - mean:
    1.9010    2.0197

Kovarianční matice GNSS senzoru - R:
    0.2316    0.0143
    0.0143    0.2170
```

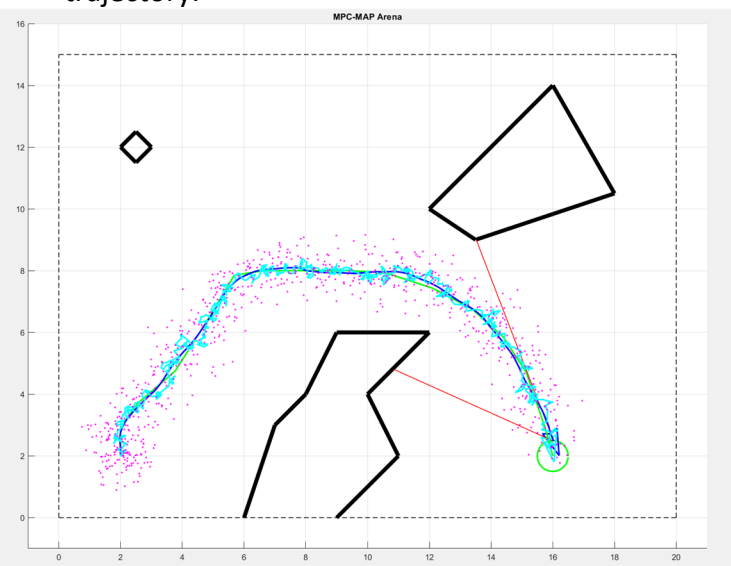
## Task 2

In the prediction step, the nonlinear motion model was implemented.
G = [1,0,-v*dt*sin(θ); 0,1,v*dt*cos(θ); 0,0,1];

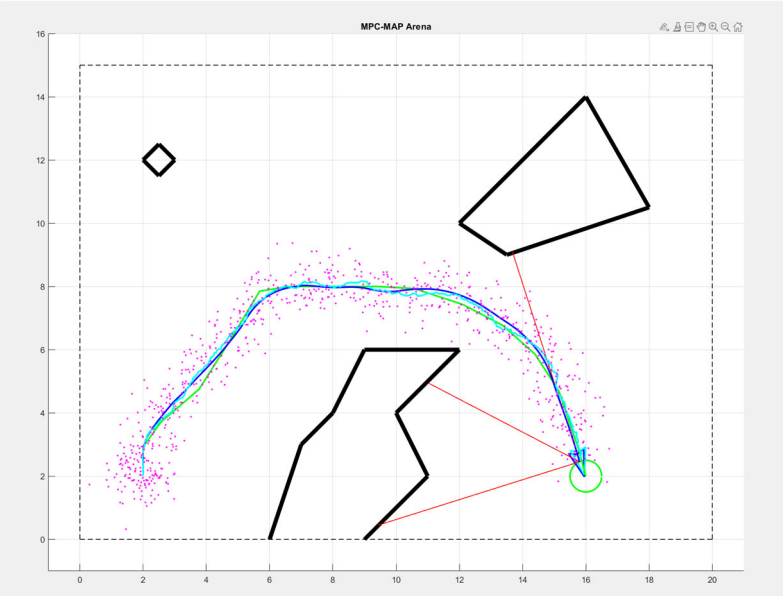
In the correction step, Linear KF is used.

## Task 3

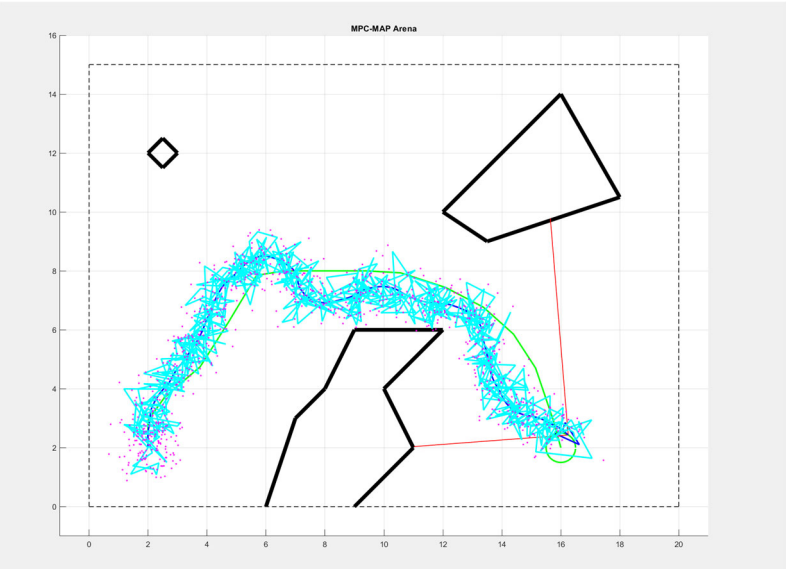
I tested three different values of the R matrix:
0.01 – Works well. Good balance between the model and GNSS; smooth and accurate trajectory.




0.0001 – Too small. Overtrusts the model, ignores GNSS; smooth but drifts over time. But also works

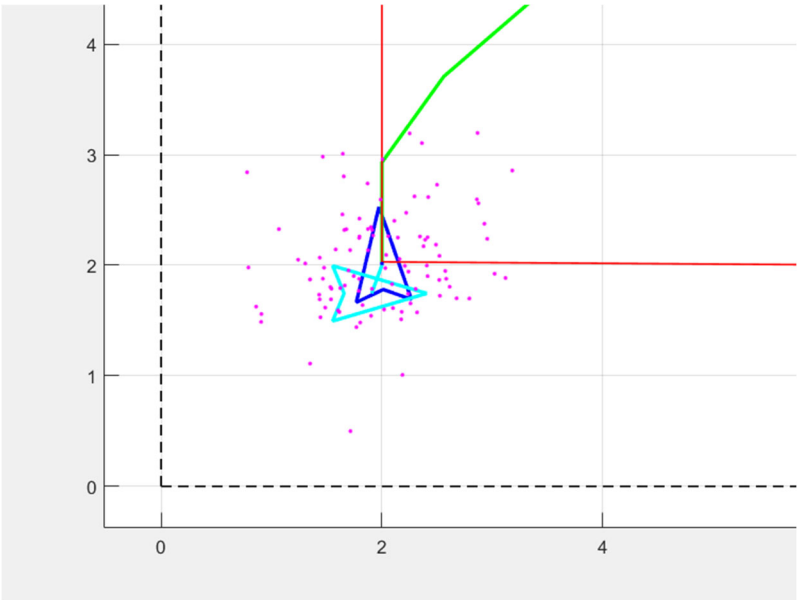


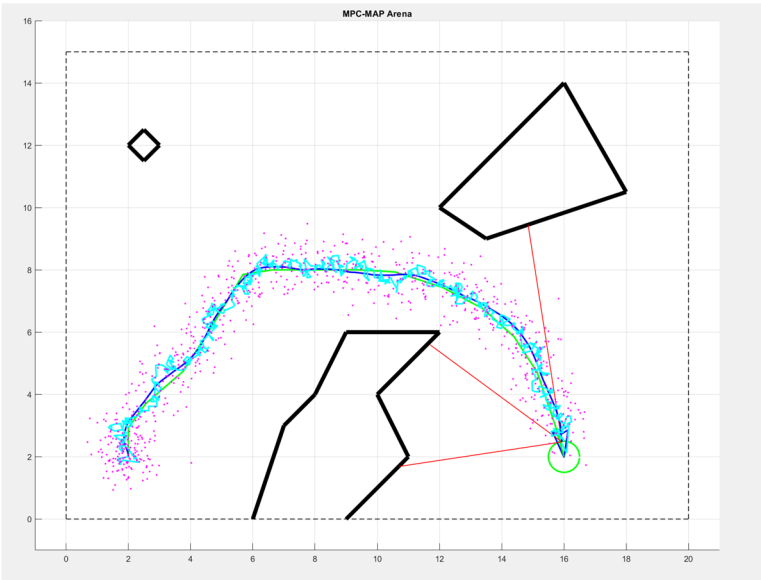

1.0 – Too large. Overtrusts GNSS; trajectory is noisy and erratic.




## Task 4

Start of simulation: 100 measurements were taken and averaged, so the X and Y position are quite reliable. The rotation is unknown, so the initial angle is set to 0° with low confidence. After a few iterations, once the robot starts moving, the rotation is gradually estimated.




End of simulation: